

Research Notes on Named entity recognition

Named entity recognition

>> Section 1

Formal evaluation To evaluate the quality of an NER system's output, several measures have been defined. The usual measures are called precision, recall, and F1 score. However, several issues remain in just how to calculate those values. These statistical measures work reasonably well for the obvious cases of finding or missing a real entity exactly; and for finding a non-entity. However, NER can fail in many other ways, many of which are arguably "partially correct", and should not be counted as complete success or failures. For example, identifying a real entity, but:

>> Section 2

with fewer tokens than desired (for example, missing the last token of "John Smith, M.D.") with more tokens than desired (for example, including the first word of "The University of MD") partitioning adjacent entities differently (for example, treating "Smith, Jones Robinson" as 2 vs. 3 entities) assigning it a completely wrong type (for example, calling a personal name an organization) assigning it a related but inexact type (for example, "substance" vs. "drug", or "school" vs. "organization") correctly identifying an entity, when what the user wanted was a smaller- or larger-scope entity (for example, identifying "James Madison" as a personal name, when it's part of "James Madison University"). Some NER systems impose the restriction that entities may never overlap or nest, which means that in some cases one must make arbitrary or task-specific choices. One overly simple method of measuring accuracy is merely to count what fraction of all tokens in the text were correctly or incorrectly identified as part of entity references (or as being entities of the correct type). This suffers from at least two problems: first, the vast majority of tokens in real-world text are not part of entity names, so the baseline accuracy (always predict "not an entity") is extravagantly high, typically >90%; and second, mispredicting the full span of an entity name is not properly penalized (finding only a person's first name when his last name follows might be scored as $\frac{1}{2}$ accuracy). In academic conferences such as CoNLL, a variant of the F1 score has been defined as follows:

>> Section 3

Jurafsky, Daniel; Martin, James H. (2008). "13.5 Partial Parsing". *Speech and Language Processing* (2nd ed.). Upper Saddle River, N.J.: Prentice Hall. ISBN 978-0131873216. Jurafsky, Daniel; Martin, James H. (2008). "22.1. Named Entity Recognition". *Speech and Language Processing* (2nd ed.). Upper Saddle River, N.J.: Prentice Hall. ISBN 978-0131873216.

>> Section 4

[Jim]Person bought 300 shares of [Acme Corp.]Organization in [2006]Time. In this example, a person name consisting of one token, a two-token company name and a temporal expression have been detected and classified.

>> Section 5

GATE supports NER across many languages and domains out of the box, usable via a graphical interface and a Java API. OpenNLP includes rule-based and statistical named-entity recognition. spaCy features fast statistical NER as well as an open-source named-entity visualizer. Hand-crafted grammar-based systems typically obtain better precision, but at the cost of lower recall and months of work by experienced computational linguists. Statistical NER systems typically require a large amount of manually annotated training data. Semisupervised approaches have been suggested to avoid part of the annotation effort. In the statistical learning era, NER was usually performed by learning a simple linear regression model on engineered features, then decoded by a bidirectional Viterbi algorithm. Some commonly used features include:

>> Section 6

Difficulties NER can have reference resolution ambiguities where the same name can refer to different entities of the same type. For example, "JFK" can refer to the former president or his son. The same name can refer to completely different types. "JFK" might refer to the airport in New York. "IRA" can refer to Individual Retirement Account or International Reading Association. This can be caused by metonymy. For example, "The White House" can refer to an organization instead of a location.

>> Section 7

Precision is the number of predicted entity name spans that line up exactly with spans in the gold standard evaluation data. I.e. when [Person Hans] [Person Blick] is predicted but [Person Hans Blick] was required, precision for the predicted name is zero. Precision is then averaged over all predicted entity names. Recall is similarly the number of names in the gold standard that appear at exactly the same location in the predictions. F1 score is the harmonic mean of these two. It follows from the above definition that any prediction that misses a single token, includes a spurious token, or has the wrong class, is a hard error and does not contribute positively to either precision or recall. Thus, this measure may be said to be pessimistic: it can be the case that many "errors" are close to correct, and might be adequate for a given purpose. For example, one system might always omit titles such as "Ms." or "Ph.D.", but be compared to a system or ground-truth data that expects titles to be included. In that

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case, every such name is treated as an error. Because of such issues, it is important actually to examine the kinds of errors, and decide how important they are given one's goals and requirements. Evaluation models based on a token-by-token matching have been proposed. Such models may be given partial credit for overlapping matches (such as using the Intersection over Union criterion). They allow a finer grained evaluation and comparison of extraction systems.

>> Section 8

Early work in NER systems in the 1990s was aimed primarily at extraction from journalistic articles. Attention then turned to processing of military dispatches and reports. Later stages of the automatic content extraction (ACE) evaluation also included several types of informal text styles, such as weblogs and text transcripts from conversational telephone speech conversations. Since about 1998, there has been a great deal of interest in entity identification in the molecular biology, bioinformatics, and medical natural language processing communities. The most common entity of interest in that domain has been names of genes and gene products. There has been also considerable interest in the recognition of chemical entities and drugs in the context of the CHEMDNER competition, with 27 teams participating in this task. In 2001, research indicated that even state-of-the-art NER systems were brittle, meaning that NER systems developed for one domain did not typically perform well on other domains. Considerable effort is involved in tuning NER systems to perform well in a new domain; this is true for both rule-based and trainable statistical systems. As of 2007, state-of-the-art NER systems for English produce near-human performance. For example, the best system entering MUC-7 scored 93.39% of F-measure while human annotators scored 97.60% and 96.95%.